

Learning Li-ion Battery Health and Degradation Modes from Data with Aging-Aware Circuit Models

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ABSTRACT

Non-invasive estimation of Li-ion battery state-of-health from operational data is valuable for battery applications but remains challenging. Pure model-based methods may suffer from inaccuracy and long-term instability of parameter estimates, whereas pure data-driven methods rely heavily on training data quality and quantity, limiting generality when extrapolating to unseen cases. This work applies an aging-aware equivalent circuit model for health estimation, combining the flexibility of data-driven techniques within a model-based approach. A simplified electrical model with voltage source and resistor incorporates Gaussian process regression to learn capacity fade over time and the dependence of resistance on operating conditions and time. The approach was validated on two datasets and shown to give accurate performance with less than 1% relative root mean square error in capacity and less than 2% mean absolute percentage error. Changes in the open-circuit voltage versus state-of-charge function strongly influence the learned resistance, enabling estimation of in-operando differential voltage curves from operational data.

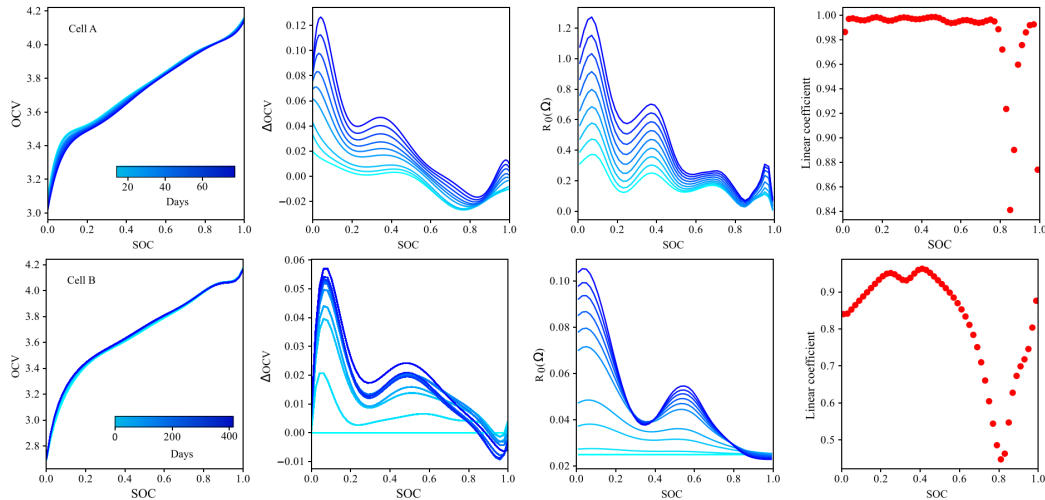


Figure 1: Aging-Aware Circuit Model for Battery Health Estimation.